1 Climate-Model Application

This chapter connects the mathematical and numerical parts of the thesis to a climate-model workflow. Its purpose is not to change the main topic of the thesis, but to test whether the methods studied in the previous chapters remain useful in a realistic downstream task.

In this setting, we no longer work only with synthetic matrices whose properties we control directly. The relevant operators come from the statistical post-processing of climate simulations. Therefore, the geometry of the problem is now shaped by the application itself: by dependence between variables, spatial correlations, temporal structure, and the scale of the data.

The main question is whether randomized and mixed-precision methods remain useful in this setting, and if so, in what sense. Usefulness should be interpreted broadly. A method may be valuable because it reduces runtime, lowers memory use, allows larger problems to be treated on the same hardware, or reveals a numerically stable regime that was not obvious from synthetic tests alone.

From the point of view of the thesis, this chapter is a test of transfer. The earlier chapters describe least-squares geometry, embeddings, reduced operators, conditioning, and floating-point perturbations. Here we ask how much of that picture survives in a genuine climate-data workflow. The answer need not be a dramatic speedup. It is equally useful if the application shows where the numerical assumptions behind the randomized or mixed-precision methods are realistic and where they begin to fail.

1.1 Multivariate Bias Correction

The relevant climate-facing context is multivariate bias correction. Univariate methods such as quantile-delta mapping (QDM) are designed to correct marginal distributions while preserving modeled changes in quantiles [1]. They are useful, but they do not by themselves repair the dependence structure across variables, locations, or times. That second task is the main reason why multivariate methods are needed.

Existing methods already separate this problem into marginal and dependence components in different ways. MBCp and MBCr combine marginal adjustment with dependence correction, while MBCn pushes further toward matching the full multivariate distribution [2] [3]. Rank-based approaches such as R2D2 extend the same line of thought to higher-dimensional dependence adjustment [4] [5]. At the same time, the climate literature emphasizes that bias correction should not be treated as a black-box distributional repair detached from physical interpretation or dynamical structure [6]. Recent comparative studies [7] have made this point quantitatively: different multivariate methods preserve different aspects of the joint distribution, and no single method dominates across all dependence metrics.

QDM, MBCp, MBCr, MBCn, and R2D2

It is worth stating explicitly how the main existing methods differ in their treatment of marginals and dependence, because the thesis proposal is best understood by comparison with these baselines.

Quantile-delta mapping (QDM) [1] is a univariate method. It preserves modeled changes in quantiles while adjusting the marginal distribution to match the observational reference. QDM does not modify dependence between variables: each column is corrected independently. In the workflow proposed in this thesis, QDM plays precisely this role — it handles the marginal step, and dependence is left to the subsequent covariance-transport layer.

MBCp [2] adds a dependence-correction step on top of marginal adjustment. It matches the Pearson correlation structure of the reference distribution while preserving the corrected marginals via rank rearrangement. MBCp is therefore a natural reference for the present thesis, because the covariance-transport map also targets second-order dependence. The key difference is that MBCp adjusts correlations directly, while the proposed method adjusts the full covariance operator through matrix square roots and inverse square roots. This gives the covariance-transport map a different algebraic structure and a natural entry point for randomized low-rank approximation.

MBCr [2] is a variant that matches the ratio of ranked values rather than correlations, and is therefore more robust to non-Gaussian features such as those in precipitation. MBCr is relevant here as a reminder that the Gaussianized latent space used in the thesis may not capture all relevant dependence structure, especially for bounded or heavy-tailed variables.

MBCn [3] is the most ambitious of the Cannon family. It iteratively applies random rotations and marginal adjustments until the full multivariate distribution of the corrected sample matches the reference. MBCn does not target covariance alone; it aims at distributional equivalence in all moments. This makes it more powerful than a covariance-only method, but also more expensive and harder to analyze numerically. The thesis does not propose to replace MBCn. The proposal is narrower: a cheaper and more analyzable covariance-based alternative in regimes where full distributional matching is either unnecessary or numerically unstable.

R2D2 [5] is a rank-resampling approach that preserves the dependence structure of the reference by resampling from its rank distribution while inserting the QDM-corrected marginal quantiles. R2D2 is conceptually simple and does not require a parametric model of dependence. It is, however, limited by the availability and smoothness of the empirical rank copula and does not naturally admit a low-rank or mixed-precision compression.

The method proposed in this thesis therefore occupies a specific position in this landscape: it is a second-order dependence-correction method, like MBCp, but it uses a regularized covariance-transport map instead of direct correlation matching, and it is explicitly designed for randomized low-rank approximation and mixed-precision perturbation analysis. It should be evaluated against these baselines not on the basis of universal superiority, but on the basis of whether its numerical-algebraic structure provides a useful tradeoff between accuracy, cost, and analyzability.

1.2 Marginals and Dependence

The proposed workflow in this thesis follows the same broad decomposition, but in a form chosen to expose the numerical linear algebra more clearly. Marginal correction is handled first by a QDM-type step. Dependence correction is then treated separately in a latent standardized or rank-normal space, where the central operator is a regularized covariance transport map of the form analyzed in ??→p.??.

This decomposition matters conceptually. It allows the climate-facing workflow to be described as

$$\text{marginal correction} + \text{dependence correction},$$

with the second term becoming the main mathematical object of the thesis. In this way, the application chapter stays tied to multivariate climate post-processing without forcing the entire theory to become a direct analysis of a fully nonlinear bias-correction algorithm.

1.3 Data

The application uses the CCCma dataset distributed with the R package `MBC` [8]. This dataset contains daily values of precipitation, temperature, and sea-level pressure from the Canadian Centre for Climate Modelling and Analysis (CCCma) global climate model, together with a corresponding observational reference. The data are organized into a historical calibration period and a future projection period, which is precisely the structure required by the bias-correction workflow:

$$X_h \in \mathbb{R}^{n\times p}, \qquad X_f \in \mathbb{R}^{m\times p}, \qquad Y_h \in \mathbb{R}^{n\times p}.$$

Here the column dimension p encodes the multivariate state. In the simplest configuration, each column corresponds to a single physical variable at a single location. More elaborate configurations may concatenate several variables, nearby grid points, or lagged features into one multivariate vector, thereby increasing p and making the covariance estimation problem dimensionally harder.

The CCCma dataset is modest in size compared to full CMIP6 archives [9], but it has the right structure for a first application-facing test: the historical and future periods are already separated, the observational reference is provided, and the number of variables is large enough to make a low-rank approach potentially interesting. In particular, the dependence between precipitation and the thermodynamic variables is known to be poorly captured by GCMs, which gives the covariance-transport step a concrete target [7].

It must be noted, however, that the basic CCCma configuration includes only precipitation, temperature, and sea-level pressure, giving $p = 3$. This is too small for a meaningful low-rank regime: with only three columns, the full covariance is a 3×3 matrix whose randomized low-rank approximation offers no computational advantage. The dataset is therefore useful as a first functional test of the pipeline — verifying that the transport map can be built and applied without error — but not as a demonstration of the computational benefit of randomized low-rank approximation. To make the low-rank setting meaningful, one must construct larger multivariate states by combining variables, locations, lags, seasons, or spatial neighborhoods, thereby increasing p and making the covariance estimation problem dimensionally harder.

For the present thesis, the dataset is used in two ways. First, it provides the real-data input for the full `randMBC` pipeline. Second, it informs the design of the climate-shaped synthetic data used in ?? → p.??: the spectral profiles and block-dependence structure of the real data are used to calibrate the synthetic generators so that the controlled experiments remain application-relevant.

1.4 Workflow

At the method level, the climate-facing procedure is described by explicit inputs and outputs. The input is the triple X_h, X_f, Y_h defined above. The output is a corrected future sample

$$X_f^{\mathrm{corr}}$$

that inherits the target marginal behavior together with an adjusted dependence structure.

The computational steps are:

1. Apply QDM-type marginal correction to obtain $\widetilde{X}_h$ and $\widetilde{X}_f$.
2. Transform the corrected calibration and future samples to a latent rank-normal space via plotting-position ranks and the inverse standard normal map.

3. Estimate the calibration covariance operators $C_X = \frac{1}{n} Z_X^\top Z_X$ and $C_Y = \frac{1}{n} Z_Y^\top Z_Y$.

4. Approximate them by randomized low-rank sketches $C_X \approx Q_X B_X Q_X^\top$ and $C_Y \approx Q_Y B_Y Q_Y^\top$, as described in ??→p.??.

5. Build the regularized transport map $T_\lambda = (C_X + \lambda I)^{-1/2}(C_Y + \lambda I)^{1/2}$ from the low-rank surrogates.

6. Apply the map to the future latent data: $Z_f^{\mathrm{corr}} = Z_f T_\lambda$.

7. Restore the target marginals by rank rearrangement: for each variable, sort the QDM-corrected values $\widetilde{X}_f(:, j)$ and rearrange them according to the ranks of $Z_f^{\mathrm{corr}}(:, j)$.

This explicit workflow is useful because it makes clear which part of the method is meant to carry the numerical linear algebra: the covariance approximation and transport layer (steps 3–6), not the entire post-processing pipeline at once.

A few remarks on the latent-space construction and rank restoration are in order. In step 2, both the historical model data $\widetilde{X}_h$ and the reference data Y_h are Gaussianized using their own empirical marginal distributions: the rank transform for $\widetilde{X}_h$ is computed from the ranks of $\widetilde{X}_h$ itself, and similarly for Y_h. The future sample $\widetilde{X}_f$ is Gaussianized using its own ranks, which is natural because the reference distribution for the future period is not available. This choice means that the latent future scores Z_f inherit the rank structure of the QDM-corrected marginals.

In step 7, the rank rearrangement restores the QDM-corrected marginal quantiles exactly. Because only the ranks of Z_f^{corr} are used, the transport-map perturbation changes the ordering of the corrected values but not their magnitudes. This is why rank restoration preserves QDM-corrected marginals by construction: the set of values in each column of $\widetilde{X}_f$ is unchanged; only their assignment to time indices is altered. The dependence correction is therefore confined to the copula structure, leaving the marginal distributions intact.

The local R package `randMBC` implements precisely this decomposition. It exposes a deterministic marginal-correction and rank-restoration wrapper around a randomized covariance-transport core. The main entry point `fit_rand_mbc` accepts the data triple together with algorithm parameters (rank, oversampling, sketch type, working precision, and regularization), and returns the corrected future sample together with operator-level diagnostics (covariance error, transport-map error, condition estimates, and runtime).

1.5 Randomized Covariance Transport

The thesis does not propose that covariance transport replaces methods such as MBCn. Its narrower proposal is a covariance-based dependence correction step that can be compressed by randomized low-rank approximation and analyzed in mixed precision. The attraction of this approach is not only speed. It is also that the operator

$$T_\lambda = (C_X + \lambda I)^{-1/2}(C_Y + \lambda I)^{1/2}$$

has a perturbation structure that can be studied directly in terms of conditioning, regularization, randomized approximation error, and floating-point error.

The main theoretical statement from ??→p.?? shows that, under suitable spectral-gap assumptions, the transport perturbation satisfies

$$\|T_\lambda - \widehat{T}_\lambda\|_2 \lesssim K_\lambda(C_X, C_Y)(\eta_A + \eta_B),$$

where η_A and η_B bound the regularized covariance perturbations and K_λ deteriorates as the regularized spectra approach zero. The climate application gives this abstract statement a concrete meaning: η_A and η_B now reflect the quality of a randomized low-rank approximation to the empirical covariance of actual climate data, and the conditioning of the regularized transport is determined by the spectral structure of that data.

From the application point of view, the proposal should therefore be interpreted as a computationally cheaper and more analyzable dependence-correction layer. It is well suited to regimes in which the dominant challenge is to modify covariance-like dependence structure in large multivariate states. It is not intended to dominate richer multivariate distribution-matching methods in every possible setting.

1.6 Experimental Design

The climate-facing experiments are organized around the same parameter axes as the synthetic benchmarks in ??→p.??, but with the real dataset replacing the synthetic generators. The decisive sweeps vary the approximation rank, the regularization parameter λ, and the arithmetic precision, while tracking numerical and statistical diagnostics simultaneously.

Numerical diagnostics

On the numerical side, the recorded quantities are:

- Covariance approximation error $\|C - \widehat{C}\|_2$, separately for source and target.
- Transport-map error $\|T_\lambda - \widehat{T}_\lambda\|_2$.
- Induced future-data perturbation $\|Z_f T_\lambda - Z_f \widehat{T}_\lambda\|_F$.
- Condition number and minimum eigenvalue of the regularized covariance.
- Runtime and peak memory use.

These are the same diagnostics as in the synthetic benchmark, extended to real data. Their role is to determine whether the perturbation bounds from ??→p.??

remain predictive when the covariance operators are no longer controlled by construction.

Statistical diagnostics

On the statistical side, the recorded quantities are:

- Marginal quantile error at selected quantile levels.
- Pearson and Spearman correlation error between the corrected future sample and the reference.
- Spatial autocorrelation error when the column dimension encodes spatial structure.
- Temporal autocorrelation error at selected lags.
- Energy score or similar multivariate discrepancy measure.

The distinction between the two groups is important. A method may show a visible operator perturbation while still producing acceptable application-level dependence corrections, or conversely may appear cheap and accurate in low-rank terms while failing to preserve the dependence features that matter downstream. The experimental design keeps both scales visible.

Reference methods

The randomized covariance-transport method is compared against two reference levels. First, the deterministic full covariance-transport baseline, in which the regularized transport map is built from the exact empirical covariances without randomized compression. Second, selected methods from the vendored `MBC` package, specifically MBCp, MBCr, and MBCn, applied to the same data triple with their default parameter settings. These reference runs do not use randomized low-rank approximation, but they provide the standard to which the climate community would compare any new method.

Parameter sweeps

The main sweeps are:

1. **Rank sweep.** Fix $\lambda = 10^{-2}$ and double precision; vary the target rank from small values up to the full column dimension p. This tests how quickly the covariance approximation error decreases and whether the transport error follows the prediction of **??** → p. ??.
2. **Regularization sweep.** Fix the rank at a moderately large value and double precision; vary λ from 10^{-6} to 10^{-1}. This tests how strongly the inverse-square-root conditioning affects the transport error and the downstream statistical diagnostics.

3. **Precision sweep.** Fix the rank and λ; compare double and single precision in the sketch products. This is the direct test of the balancing-inexactness principle from ??→p.??: does single precision remain acceptable because randomized truncation already dominates, or does arithmetic error become visible?

4. **Basis-mode comparison.** For the same rank, λ, and precision, compare the independent-subspace and joint-basis variants described in ??→p.??. The synthetic benchmark already showed that the joint basis can dramatically reduce transport error; the climate experiment tests whether this advantage survives on real covariance operators.

1.7 Results Framework

The climate-facing experiments have been designed and the software infrastructure is in place, but the full numerical results are not yet available. The following describes the expected structure of the results and the interpretive framework within which they should be read.

Expected rank–error behavior

If the empirical covariance operators of the climate data exhibit spectral decay — as one would expect for spatially correlated fields — then increasing the approximation rank should systematically reduce the covariance error. The transport error should follow, but with the amplification predicted by K_λ. In particular, if the regularized minimum eigenvalue remains well above machine precision, the transport error should decrease at a rate similar to the covariance error, up to the conditioning constant. If the regularization is too weak, the inverse square root will amplify the covariance perturbation, and the transport error may remain large even when the covariance error is already small.

Expected precision behavior

In the regime where randomized truncation error dominates arithmetic error, double and single precision should produce nearly identical results. This is the balanced regime

$$\eta_{\mathrm{fp}} \lesssim \eta_{\mathrm{rand}},$$

advocated in ??→p.??. As the rank increases and the approximation error decreases, a crossover point should appear beyond which single precision degrades the result. The location of this crossover depends on the spectral decay and on the regularization: stronger regularization moves the crossover to larger ranks because it keeps the inverse square root in a less sensitive regime.

Expected comparison with MBC methods

The deterministic full covariance-transport baseline is expected to produce reasonable dependence corrections on the CCCma data, since the method is mathematically well-defined whenever the regularized covariance is positive definite. The comparison with MBCp, MBCr, and MBCn is less predictable a priori. MBCn, in particular, targets the full multivariate distribution and may outperform the covariance-transport method on dependence metrics that go beyond second-order structure. The value of the comparison is not to claim overall superiority, but to identify which aspects of the dependence correction the covariance-transport layer captures well and which it misses.

Interpretation

The results should be interpreted along the lines already established in the synthetic benchmark. If the randomized low-rank method produces transport errors that are small relative to the statistical variability of the data, then the computational savings are well motivated. If the transport errors are large but the downstream statistical diagnostics remain acceptable, then the operator-norm viewpoint is too strict and the application provides a meaningful relaxation. If both the operator perturbation and the statistical diagnostics are poor, then the approximation regime itself is the bottleneck, and precision considerations are secondary until the rank and regularization are brought into a workable range.

1.8 Validation Questions

The application chapter should evaluate the method at two levels. The first level is numerical: how accurately is the covariance operator approximated, how sensitive is the regularized transport map to lower precision, and how much regularization is needed to keep the inverse square root in a benign regime? The second level is statistical: after the dependence correction is pushed back to the climate data, which marginal and multivariate features are preserved, improved, or degraded?

The validation metrics are therefore split into two groups, as described in the experimental design above. These questions are deliberately broader than raw runtime. Reduced memory use, stable lower-precision behavior, and a clearer map of failure regimes are all valid outcomes. In a realistic post-processing workflow, understanding when the method fails is often as valuable as demonstrating a speedup.

1.9 Limitations

The limitations should be stated plainly. Covariance transport captures only one layer of multivariate structure. It does not by itself represent every nonlinear, tail-dependent, or temporally structured feature that a richer bias-correction method might target. It is therefore best viewed as a deliberately focused numerical core: a dependence-correction operator that is rich enough to be climate-relevant, but structured enough to admit randomized low-rank approximation and mixed-precision perturbation analysis.

A further limitation concerns the dataset. The CCCma data included with the `MBC` package is a useful first test, but it is small by the standards of current CMIP6 archives. Whether the computational advantages of the randomized approach become decisive at larger scales remains an open question that this thesis does not resolve. The present chapter should be read as a proof of concept for the transfer of the mathematical framework to real data, not as a definitive application study.

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